Match each name to its drawing.

R01



? ? ? ?

Line Point Ray Line segment

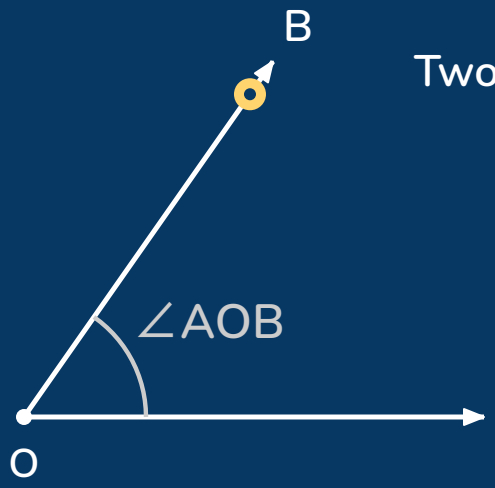
Check

R01 Point, line, segment and ray

Drag each name below a drawing. A correct match reveals its meaning.

When two rays have a common endpoint, they form an angle.

R02



Two rays, one common endpoint

Arms: OA and OB

Vertex: O

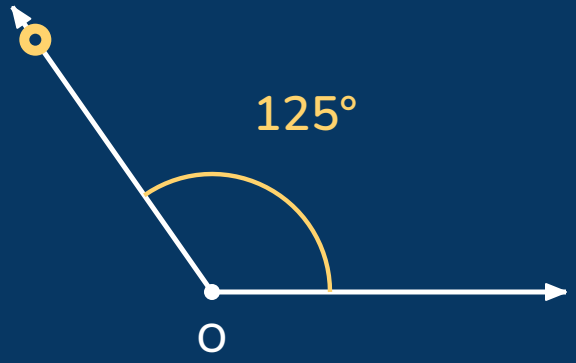
Next

R02 Angle, arms, vertex and notation

Move ray OB. Then tap the arms and the shared endpoint.

Drag the arm. Watch the angle change.

R03



Angle size uses degrees (°).

Acute	Greater than 0°, below 90°
Right	Exactly 90°
Obtuse	Between 90° and 180°
Straight	Exactly 180°
Reflex	Between 180° and 360°

Obtuse

Quarter-turn 90° Half-turn 180° Full turn 360°

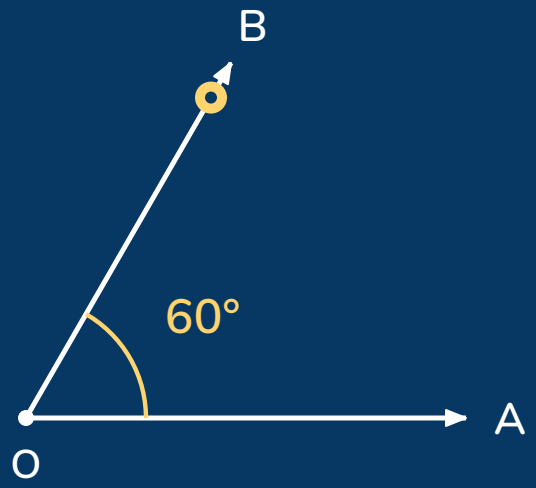
Next

R03 Angle types and degree measure

Rotate OB through a full turn. The measure and angle name update.

Make an obtuse angle.

R04



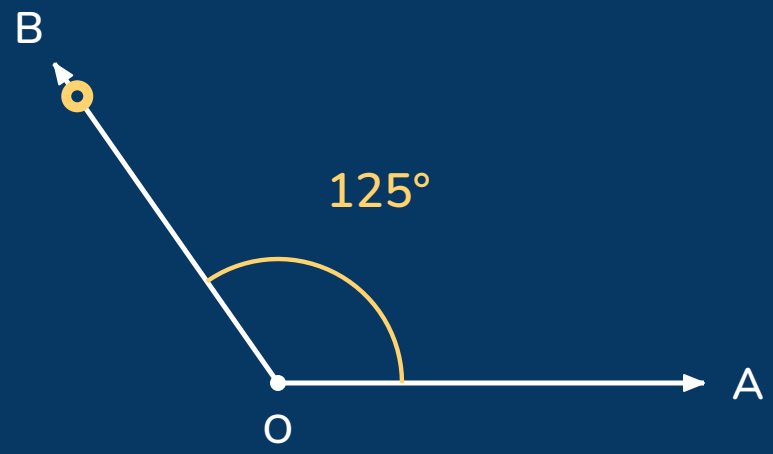
Check

R04 Recall CFU: obtuse angle

Drag OB, then tap Check. No angle-name cue appears during the attempt.

Now make a reflex angle.

R05



Check

R05 Recall CFU: reflex angle

Continue rotating OB past a straight angle, then tap Check.

Find each pair’s total.

R06

Pair A



35° + 55° = ? °

Complementary: total 90°

Pair B



125° + 55° = ? °

Supplementary: total 180°

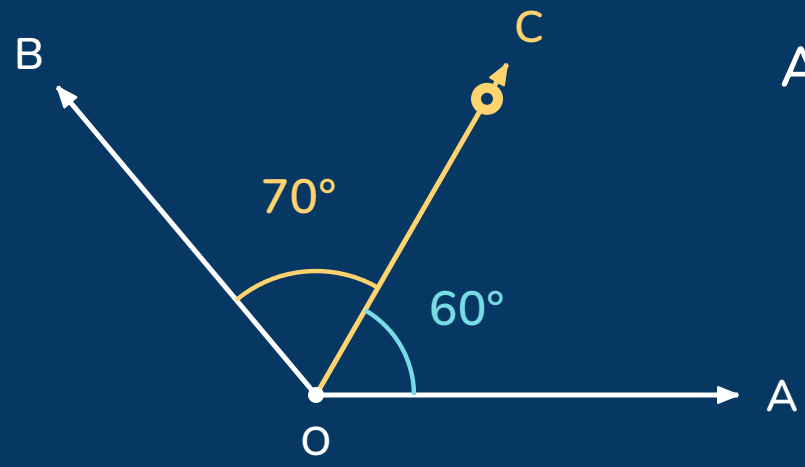
Check

R06 Complementary and supplementary angles

Add the two angle measures in each pair. The names appear after submission.

When two angles share the same vertex and have a common arm they are called adjacent angles.

F01



Adjacent angles

Next

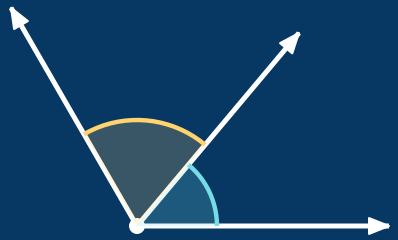
F01 Adjacent angles

Drag C between OA and OB. Tap the shared vertex and common arm.

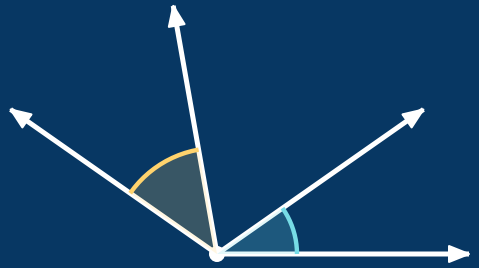
Which pair is adjacent?

F02

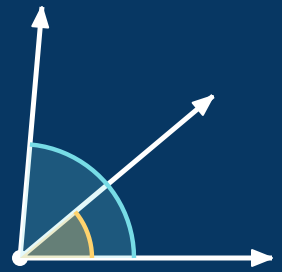
A



B



C



A B C

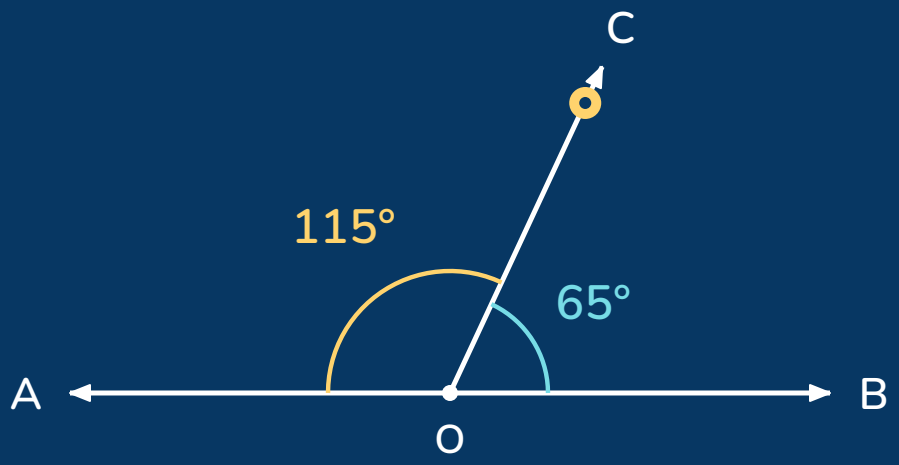
Check

F02 CFU: adjacency

Tap A, B or C, then Check. Look at the marked angle regions.

Drag the ray. What stays the same?

F03



65° + 115° = 180°

Their sum stays 180°.

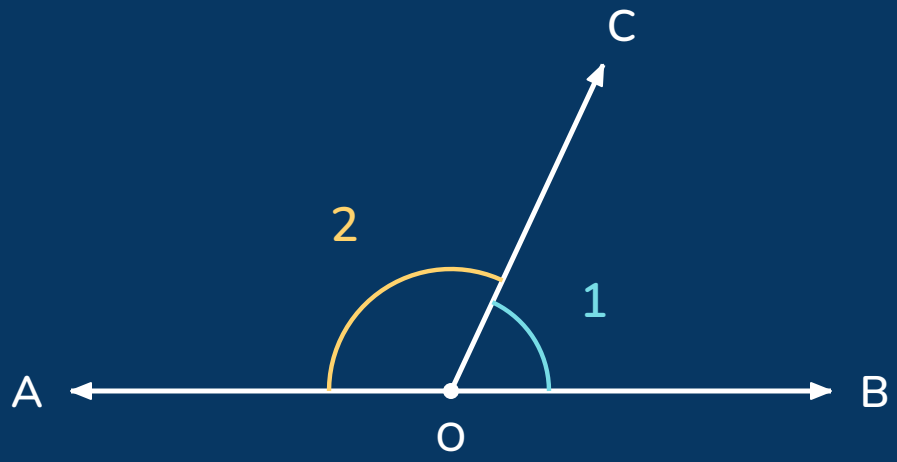
Each angle stays the same.

Next

F03 Discovery: fixed sum

Move C to several positions. Then choose the statement that remains true.

These angles form a linear pair.



Linear pair

Adjacent angles

Non-common arms form a line

Linear Pair Axiom

$\angle 1 + \angle 2 = 180°$

Next

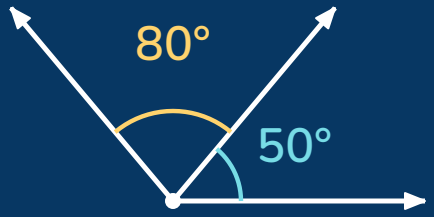
F04 Linear pair and Linear Pair Axiom

Tap the common arm, then the two non-common arms.

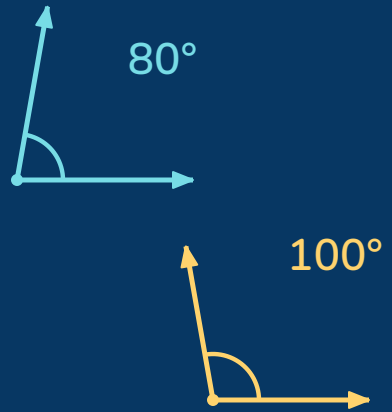
Which pair forms a linear pair?

F05

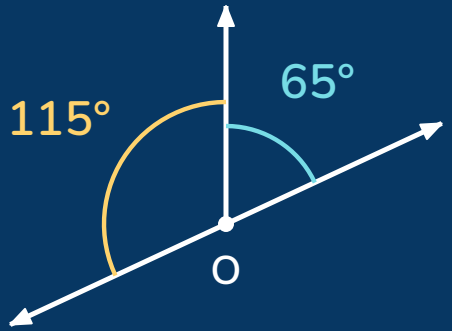
A



B



C



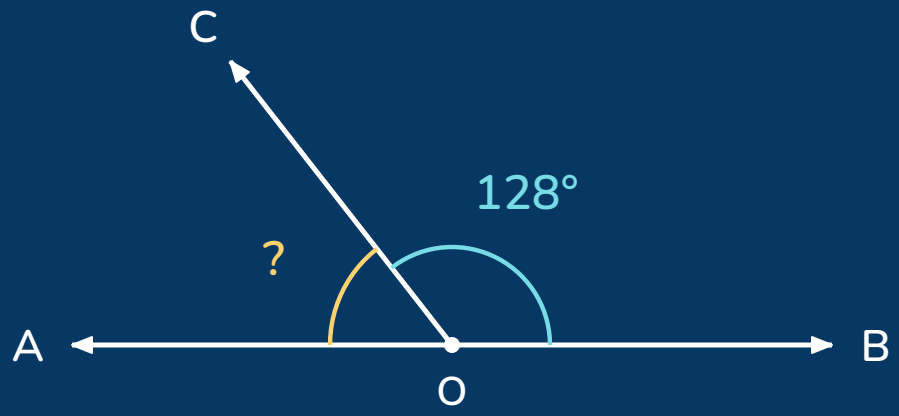
Check

F05 CFU: linear pair conditions

Tap the correct pair, then Check.

Find the missing angle.

F06



128° + ?

= 180°

Missing angle: ? °

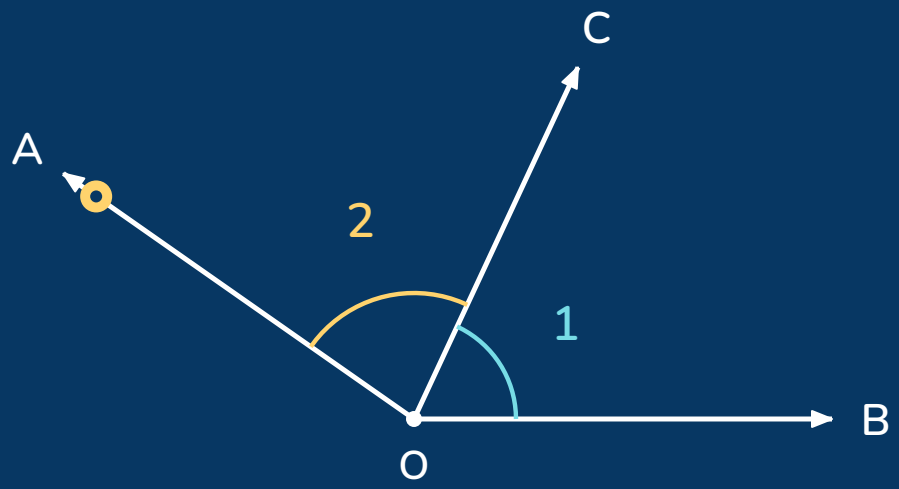
Check

F06 Guided application: unknown angle

Complete 128° + ? = 180°, then enter the missing measure.

Make a linear pair.

F07

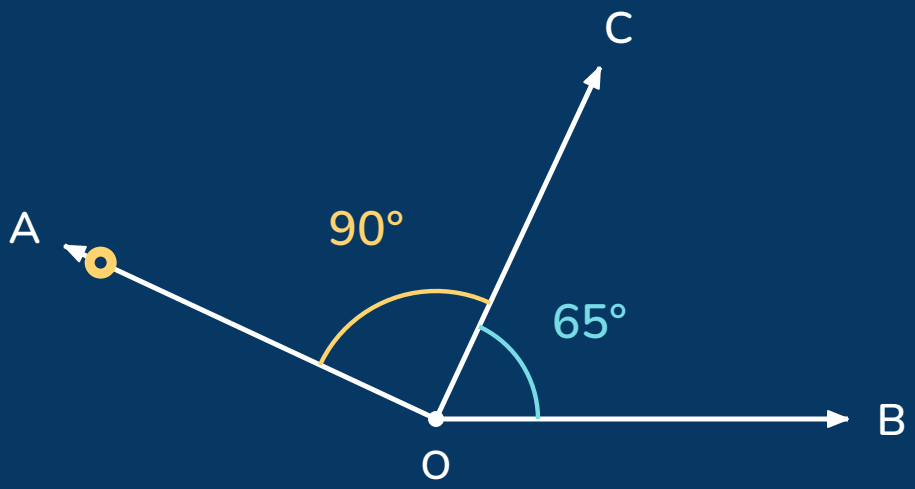


Check

F07 CFU: construction

Move A. Keep OB and OC fixed. Tap Check when your pair is ready.

Drag A until the total is 180°.



$65° + 90° = 155°$

What happens at 180°?

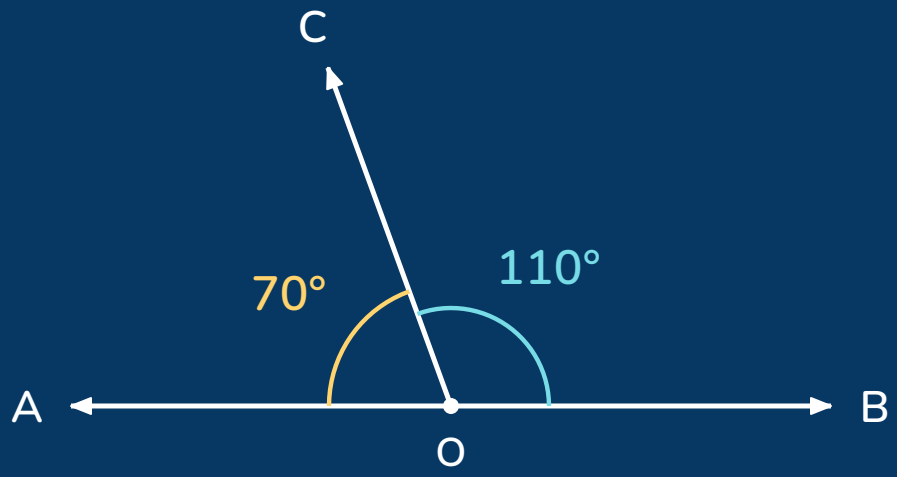
Next

F08 Converse: reverse the exploration

Move the non-common arm OA. Watch the two measures and their sum.

If adjacent angles total 180°, what follows?

F09



Their non-common arms form a line.

All three arms form a line.

Converse of the Linear Pair Axiom

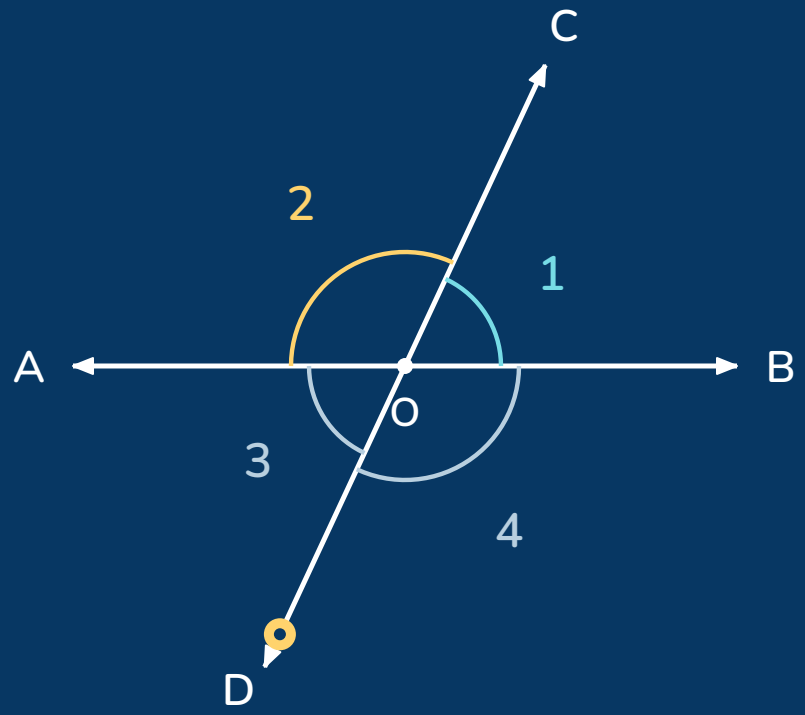
Next

F09 Converse of the Linear Pair Axiom

Choose the statement supported by the construction.

Extend the ray through O.

F10



AB and CD intersect at O.

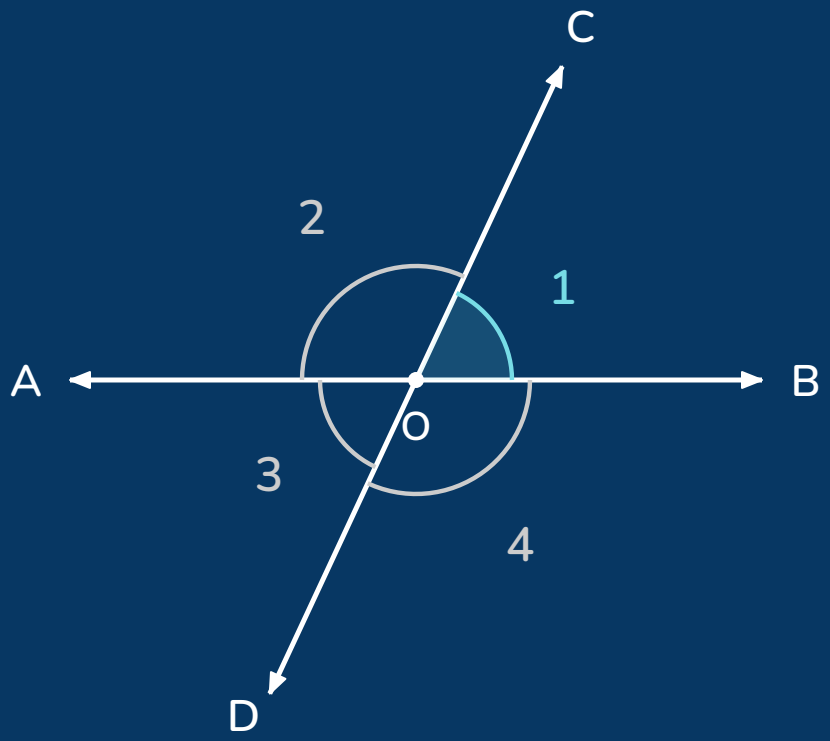
Next

F10 Intersecting lines

Drag the lower handle away from O to complete line CD.

Tap the angle opposite ∠1.

F11

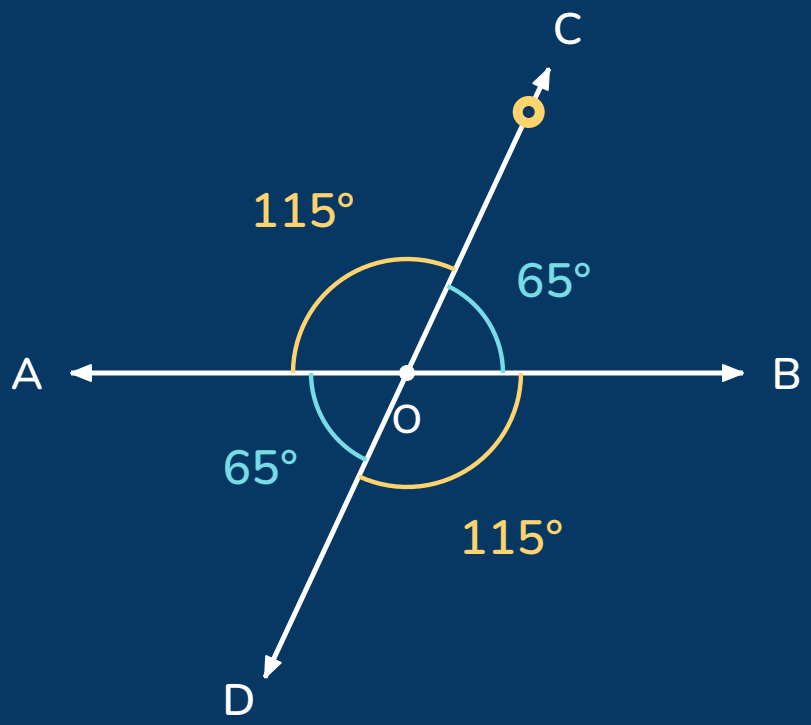


Check

F11 Vertically opposite angle pairs

Select the angle across the intersection. Then identify the other opposite pair.

Rotate CD. Compare opposite angles.



Opposite pair 1 and 3

65° = 65°

Opposite pair 2 and 4

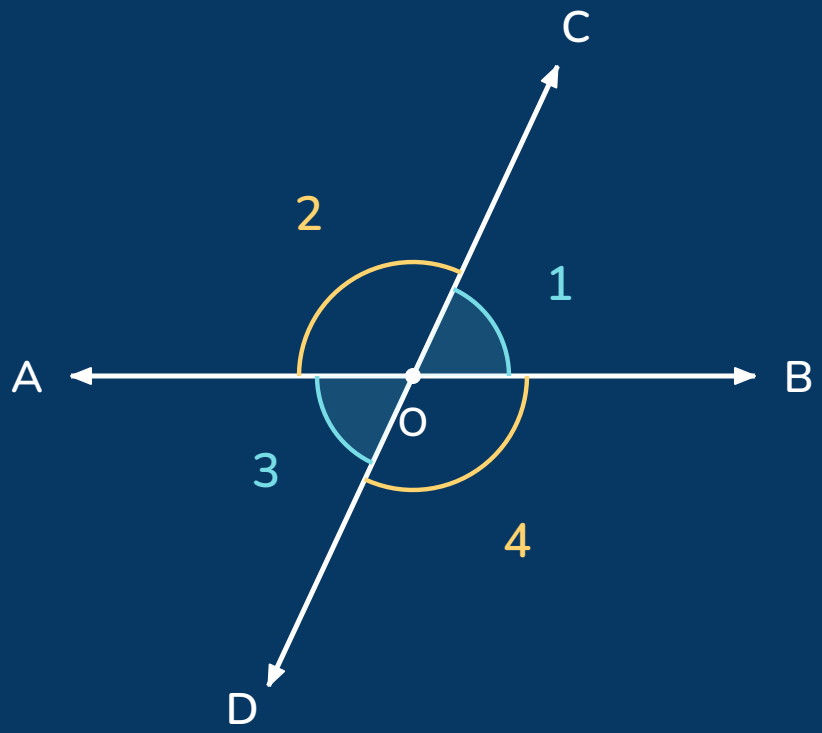
115° = 115°

Next

F12 Exploration: equal opposite angles

Rotate the complete line CD through several positions. All four measures update.

Can we show ∠1 = ∠3 without measuring?



Given

AB and CD are lines.

To show

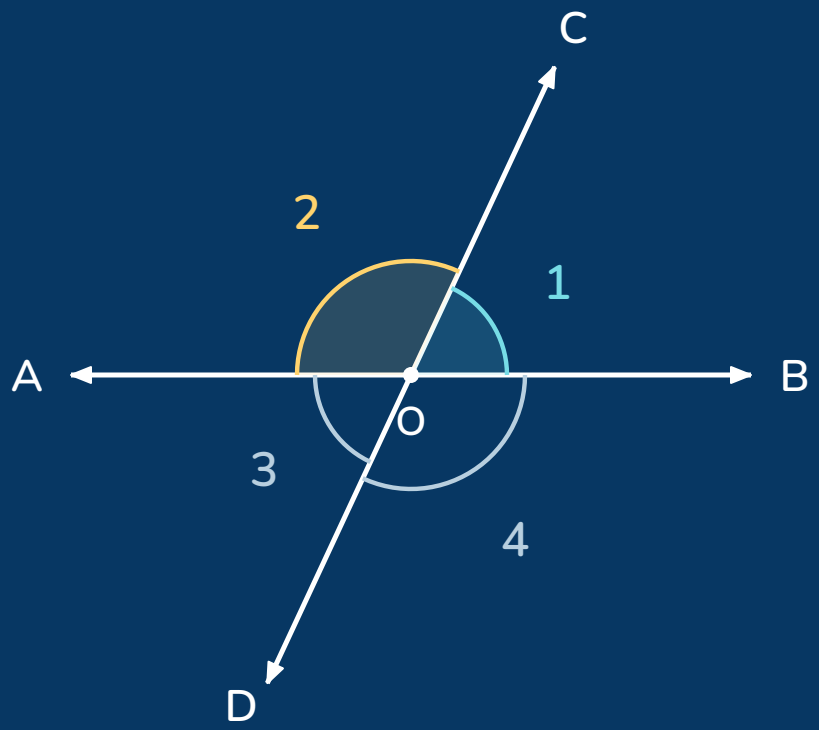
$\angle 1 = \angle 3$

Next

F13 Proof goal

Tap the two angles we want to compare.

Complete the first equation.



$\angle 1 + \angle 2 =$?

Why?

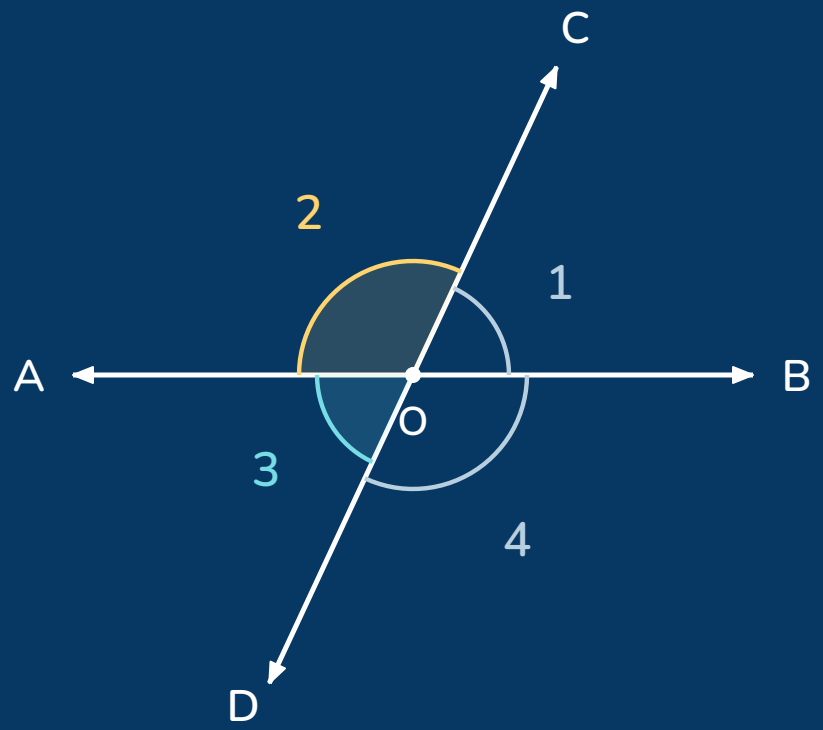
- Linear pair
- Vertically opposite
- Measured values

Check

F14 Proof step 1: first linear pair

Enter the total of $\angle 1$ and $\angle 2$. Then choose a reason.

Use ∠2 in another linear pair.



$\angle 1 + \angle 2 = 180°$

$\angle 2 + \angle 3 =$?

Reason: linear pair

Check

F15 Proof step 2: a shared supplement

Complete ∠2 + ∠3 = ?. Give the same geometric reason.

Both sums equal 180°.

$$\angle 1 + \angle 2 = 180°$$

$$\angle 2 + \angle 3 = 180°$$

$$\angle 1 + \angle 2 = \angle 2 + \angle 3$$

Next

Connect the two equal totals. Then compare the expressions.

Subtract ∠2 from both sides.

F17

$$\angle 1 + \angle 2 = \angle 2 + \angle 3$$

Subtract the same angle on both sides.

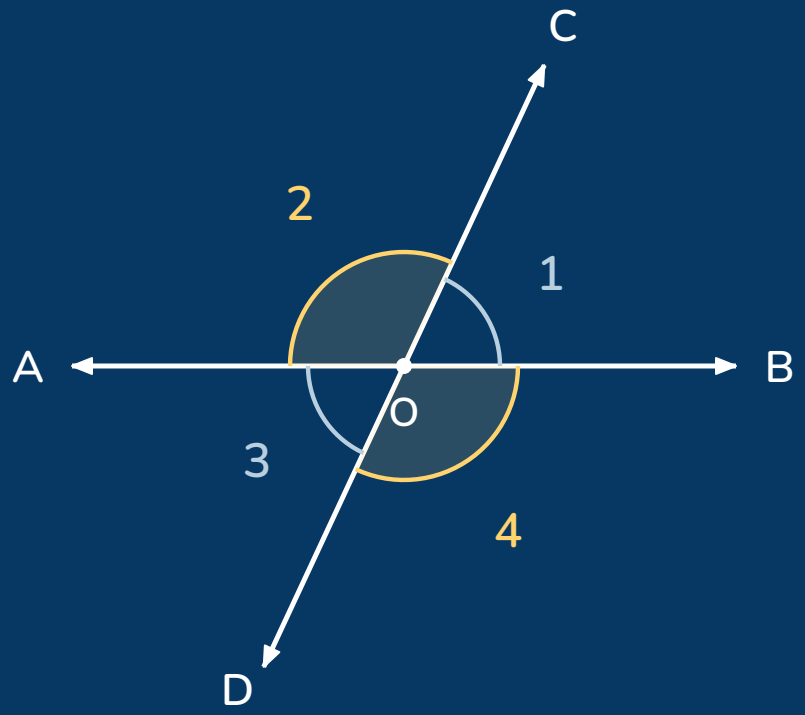
$$\angle 1 = \angle 3$$

Next

F17 Proof step 4: subtract the same angle

Tap the common ∠2 term on each side of the equation.

Now show ∠2 = ∠4.



∠2 + ? = 180°

∠4 + ? = 180°

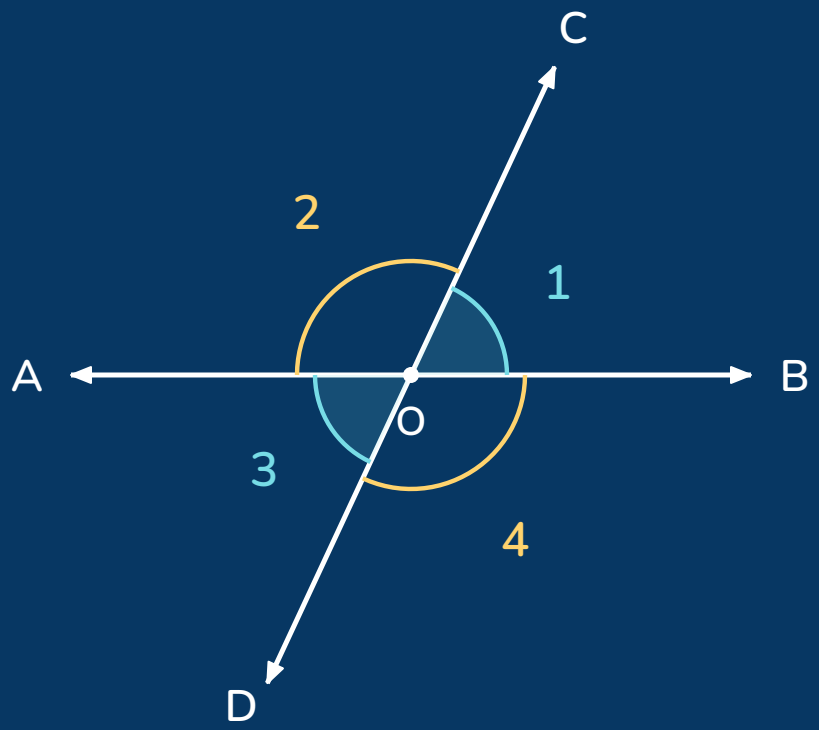
∠1 ∠3

Check

F18 CFU: prove the second opposite pair

Choose one shared angle for both linear-pair equations. Then cancel it.

When two lines intersect, opposite angles are equal.



Vertically opposite angles

$\angle 1 = \angle 3$

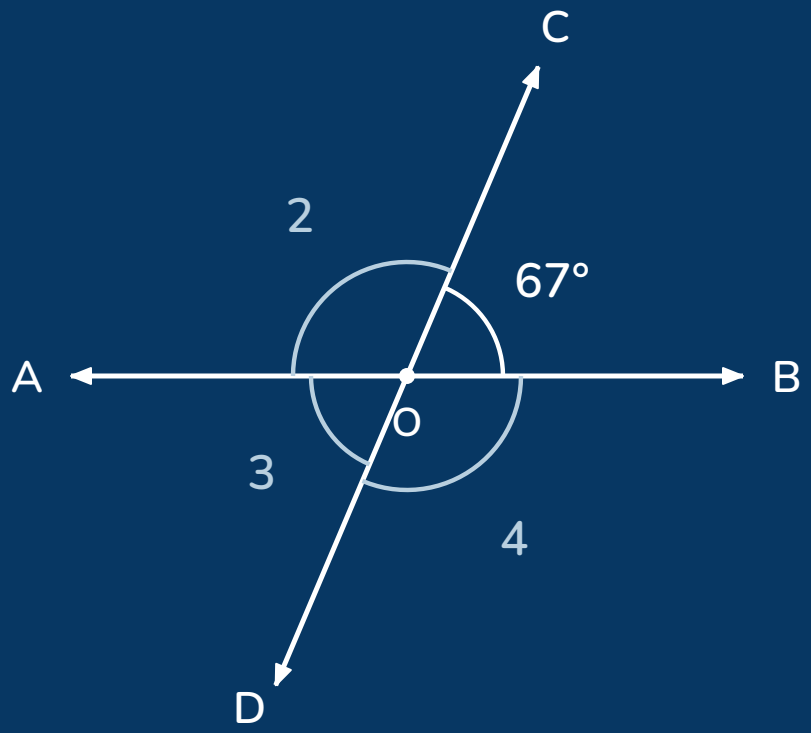
$\angle 2 = \angle 4$

Next

F19 Vertically Opposite Angles theorem

Tap each opposite pair to connect the theorem to the proof.

Find the three missing angles.



$\angle 2 =$? °

$\angle 3 =$? °

$\angle 4 =$? °

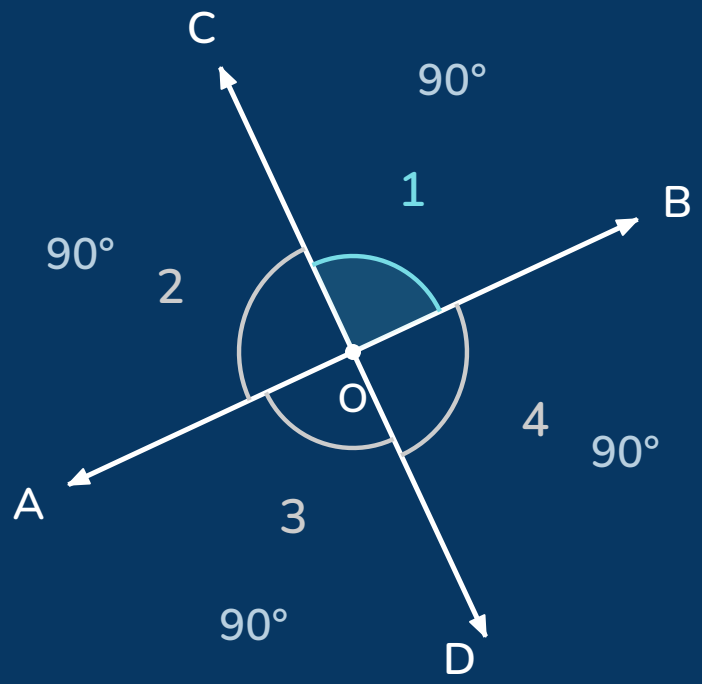
Check

F20 CFU: complete an intersection

Enter the measures of angles 2, 3 and 4. Then choose reasons.

Which angle is vertically opposite to ∠1?

F21



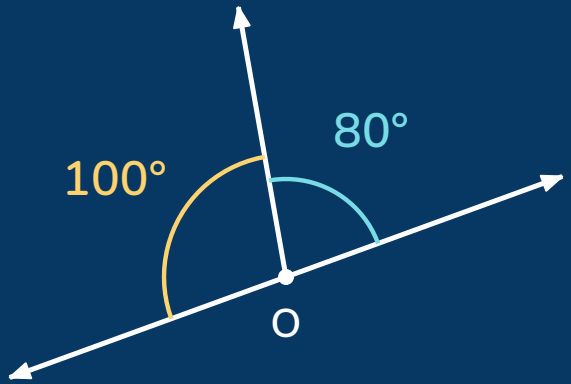
Check

F21 CFU: opposite pairs at a right-angle crossing

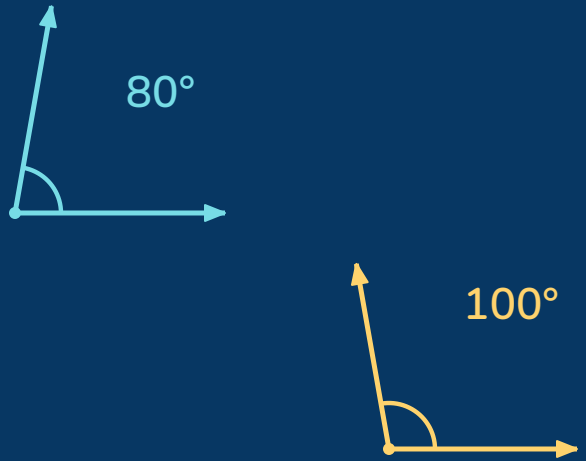
Tap the angle across the vertex, even though every angle is 90°.

Which pair’s non-common arms must form a line?

Pair A



Pair B

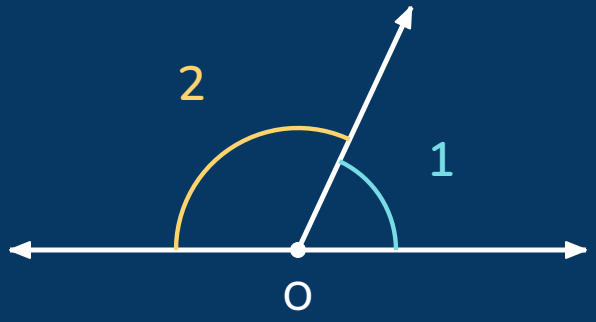


Check

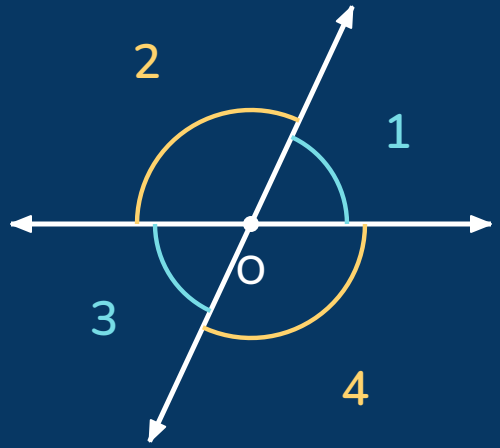
F22 CFU: the converse and its conditions

Choose a pair and explain which conditions allow the conclusion.

You can find angles without measuring them.



Linear pair: total 180°



Vertically opposite: equal

Adjacent + total 180°: the non-common arms form a line.

Next

F23 Learning close

Tap each relationship for a final recap, then finish the learning section.